Systemic Optimization — Decision Audit v0.2.1

WORKSHOP WORKSHEET · DRAFT v0.2.1 · 7 NOVEMBER 2026

Status: Exploratory decision-support instrument, **not validated**. It does not produce a risk score, optimize a policy, or establish that a real-world decision is safe or correct.

Purpose. Examine whether a decision is distorted by its objectives, causal assumptions or the evidence its own policy makes available; compare feasible remedies and their unintended consequences.

New in v0.2.1. A worked Q3 selective-evidence probe, explicit combined packages, a 45-minute mode, an option to reframe the decision, and an optional one-page after-action review. **Freeze for pilot testing after this revision.**

How to run the workshop — quick or full

Phase	Quick · 45 min	Full · 85 min	Workshop output
0. Frame decision and roles	5	8	Decision, owner, affected people, failure condition
1. Decision relevance	7	10	What plausible assumption could change the action?
2. Four probes (Q3 mandatory)	20	35	Objectives, model, blind spots, feasible learning
3. Package + coupling matrix	8	20	Comparable responses, exposure and costs
4. Decision and review record	5	12	Action or reframe, owner, safeguards, review trigger

Phase 2 allocation: Quick Q1 3 min · Q2 3 min · **Q3 10 min** · Q4 4 min. Full: **at least 15 min for Q3**. Q3 is never skipped; mark unknowns if time runs out.

Facilitation: Phase 1 starts with 3 minutes of silent individual work. Q3 also starts with 3 silent minutes. Name an assumption challenger: *"Suppose this belief is false. What could our policy still hide?"* Record disagreements, including the decision owner's objections; absent stakeholders have not consented.

Evidence key: [O] observed · **[M]** model estimate · **[H]** hypothesis · **[U]** unknown. Example-only: **[C]** confirmatory · **[D]** prespecified descriptive · **[E]** exploratory · **[S]** model-structural · **[IP]** interpretive. None is direct field validation.

Phase 0 · Decision card

Prompt	Workshop record
Decision / organization / date	_____
Decision owner; people present; designated challenger	_____
Action required, deadline and horizon	_____
Feasible alternatives, including **keep current policy**	_____
Outcomes that define success	_____
Whose interests, harms and rights count (and whose are absent)?	_____
How would we recognize that our decision was wrong?	_____
Constraints: law, ethics, budget, capacity, reversibility	_____
Baseline assumptions / current rationale	_____

Workshop principle: Compare policy **packages** using the same plausible worlds and multiple relevant outcome criteria. A package changing both a model and an objective does not isolate either one's causal effect.

Phase 1 · Does the uncertainty matter to the decision?

Individually, before discussion: Under what plausible assumptions would you choose a different action?

Scenario / contested assumption	Preferred feasible action	Evidence [O/M/H/U] and reason
Current best account	_____	_____
Strongest credible alternative	_____	_____
Challenger's alternative / outside view	_____	_____
Optional: scenario most likely to change our mind	_____	_____

Gate result: ☐ Same action across examined scenarios ☐ Different actions ☐ Cannot tell yet

What specifically reverses the decision? _____

Smallest useful sensitivity check: _____

Example: Reconsider the action with a contested parameter at its plausible extreme; or ask whether anyone would act differently if a central assumption proved false. **Cannot tell yet?** Return to Phase 0: make the decision, actions and success criteria precise enough to compare.

Mandatory safeguard: A "same action" result may justify **shortening Phase 2 Q4**, but **never skipping Q3**. The scenarios imagined by the team may themselves exclude the blind spot. "No blind spot identified" is **not proof of absence**.

Phase 2 · Four diagnostic probes

Q1 · Are we valuing the right outcomes?

- What is rewarded, penalized, omitted or treated merely as a proxy? Who chose the objective?
- Whose interests, distributional impacts, ecosystems, longer-term effects or future goals are excluded?
- If we change what is valued, **what action changes**, and what is its price even when today's assumptions are right?

Missing criterion / affected party: _____

Alternative criterion, trade-off, evidence status: _____

Q2 · Is the causal model adequate for this decision?

- **What if two assumptions are wrong together?** Could the importance of an omission reverse under parameter error, shocks or another missing mechanism?
- Which mechanism, interaction, lag, feedback, parameter or externality could be absent or wrong?
- Distinguish **assuming** a mechanism, **observing** evidence about it and **validating** the model.

Most consequential uncertainty: _____

What action changes across models? What test could disconfirm the account? _____

Phase 2 · Diagnostic probes (continued)

Q3 · What does our own policy prevent us from seeing?

Mandatory even after a "same action" gate. Everyone completes a 3-minute silent draft before discussion. The challenger assumes a key belief is false and asks: **What does our policy select into our evidence base—and what does it select out?**

Filled fictional example (not a claim about any real lender):

Belief / exclusion rule	Action	Excluded case or state	Unavailable counterevidence
Low-scoring applicants would not repay	Reject loans	Rejected applicants under this lender's loan	Their repayment *if this lender had approved*

Your decision:

Present belief / exclusion rule	Action it induces	Case, state or group not encountered	Counterevidence unavailable
_____	_____	_____	_____
_____	_____	_____	_____

- Could the policy yield **the same observations** whether the belief is true or false? What counterfactual is missing?
- Are we observing only selected people, permitted interventions, or states our actions allow to occur?
- Can independent measurements or safe comparisons help without changing treatment? **Exposure is not yet evidence.**

Blind-spot assessment: ☐ None identified (not proof) ☐ Plausible but untested ☐ Demonstrated under stated conditions

Evidence suppressed / source that could challenge us: _____

Who would bear risk or cost of opening the blind spot? (required): _____

Consent, legality, oversight, safer alternatives or reason not to probe: _____

Q4 · Can feasible learning improve the decision *in time*?

1. **Exposure:** When will the suspected mechanism, group or state first be encountered?

2. **Evidence:** When will competing hypotheses first predict **different observations**? Exposure alone does not guarantee identification.

3. **Inference and action:** Who can interpret the evidence, when, and how much time remains to change course?

4. **Alternative to inference:** Would a simple state-contingent rule—*"if X is observed, do Y"*—capture most of the benefit without identifying the causal mechanism?

5. **Decision consequence: What would we do differently if we knew the answer?** If nothing, learning may have little decision value.

6. **Net value:** Relative to keeping or hedging, is the benefit of a **feasible** learner greater than delay, money, risk and burdens imposed on others? Do not compare only with an omniscient oracle.

Proposed probe / sensor / rule	First exposure → first distinguishable evidence	Earliest actionable change	Cost, risk-bearer, safeguards
_____	_____	_____	_____

Decision-changing observation ("if X, then Y"): _____

Could a simpler rule or robust policy suffice? _____

Example of a proposed rule, not a tested result: "If wetland stock falls below threshold X, restore; otherwise hold." Compare its outcomes and costs with inference-dependent plans.

Phase 3 · Compare responses and their coupling effects

Use **one matrix** for both remedy and coupling. Each row must denote an *actual, feasible policy package*, not an isolated suggestion. The named responses are **components you may combine**. If combining components, specify them in the **combined package** row; do not assume their effects add up. Compare the same plausible worlds, horizons and outcome criteria. Label each claim **[O/M/H/U]**; write **[U]** rather than invented precision. Changing two components does not isolate either effect.

Decision objective(s), common scale(s), baseline: _____

Response / package	If current account is right	If important alternative is right	What does it enable or suppress us to observe?	Cost, risk, reversibility, other criteria
Keep current policy	_____	_____	_____	_____
Change objective / constraints	_____	_____	_____	_____
Revise model / assumptions	_____	_____	_____	_____
Probe / **observe** to create useful evidence	_____	_____	_____	_____
Hedge / flexible or reversible plan	_____	_____	_____	_____
Combined package — name components	_____	_____	_____	_____

Cross-check the causal loop: Values and beliefs → actions → encountered reality → observations → updated beliefs → later actions.

- If **Q1** changes, which actions—and therefore **Q3** observations—change?
- If **Q2** changes, could the chosen policy now conceal evidence against the new model?
- Could protecting outcomes **reduce the decision value of learning**, even while exposure becomes easier?
- Does a supposedly better-informed response sacrifice a neglected outcome, or transfer burdens to people absent from the workshop?

Most consequential coupling / unintended effect: _____

Most defensible response across the examined plausible worlds: _____

What insurance does a hedge buy, and what does it cost when the original model is right? _____

Which rows are* not *directly comparable (objectives, timing, assumptions, evidence)? _____

Example: A changed-objective package and the baseline cannot be ranked on different scores without agreeing how to weigh criteria; probing and hedging may operate over different horizons.

Phase 4 · Decision, accountability and learning record

Choose the Phase 3 row that describes the action (or use the separate reframe disposition):

☐ **Keep** ☐ **Change objective** ☐ **Revise model** ☐ **Probe / observe** ☐ **Hedge** ☐ **Combined package**

Chosen package / exact next action: _____

Why this response under both current and alternative assumptions? _____

Which outcome criteria or groups may be worse off? What safeguards apply? _____

Single most consequential unresolved assumption: _____

Smallest worthwhile evidence check—or reason to stop learning: _____

Owner, deadline and review date: _____

Trigger to reconsider / stop / falsify (if X happens, do Y): _____

Dissenting view / affected perspective not represented: _____

Whose view could most change the decision, and why are they absent? _____

Decision-owner rejection of a finding, with reason (if any): _____

Evidence ledger

Claim important to decision	[O/M/H/U]	Source, limitation, owner of next check
_____	_____	_____
_____	_____	_____
_____	_____	_____

Two-minute pilot feedback (complete even if the decision did not change)

- **Did this expose a consequential issue not previously considered?** ☐ Yes ☐ No ☐ Unsure
- **Did it alter the action or information plan?** ☐ Yes ☐ No ☐ Too early
- **Actual time spent:** _____ minutes. **What would you cut?** _____
- **Most useful question / most confusing or redundant question:** _____
- **Did anyone incur, or could anyone incur, risk from the proposed test?** _____

Disposition: ☐ Adopt provisional action ☐ Seek specific evidence ☐ **Reframe decision** ☐ Stop: further analysis not worth its cost

Important: A completed worksheet is a **record of reasoning, not a decision rule**. It is not independent validation, a substitute for affected-party participation, or proof that the framing is adequate.

Appendix A · Worked example: A2 synthetic flood management

Illustrative application, not a real catchment assessment or validation of this instrument. Sources: Working Paper 1 v0.2, §§4–7, Appendix B (T5), and its evidence-status appendix. Manuscript in the research repository.

Decision, scenarios and confirmatory package result

Three municipalities allocate flood-adaptation measures over **11 periods**. A physical retention mechanism **R** may be present or absent. The planner may include or omit R in its model. Conventional objective **U1** and ecological objective **F1-eco** value outcomes differently.

Confirmatory package	Model assumption	Objective
REP	Assume R	Minimize conventional U1
COMP	Omit R	Optimize F1-eco

[C/M] Over **40 independently drawn synthetic worlds**, with declared 50/50 hypothetical weighting of R present and absent: **REP − COMP** = **−1.112 U1** (95% CI **−1.737 to −0.487**). Negative favours REP on U1. The interval half-width, about **0.625**, **missed** the target **0.50**. COMP performed better on wetland outcomes. Because both model and objective differ, this is a *package comparison*, not an isolated estimate of adding R.

Filled diagnostic probes

Q1 · Values. [C/M] U1 and wetland criteria favor different aspects of the policies; an audit must retain both outcomes rather than collapsing the trade-off into a claim of overall superiority.

Q2 · Model. [E/M] The omission's decision relevance changes with parameter errors and other model assumptions. The observed mean U1 advantage of REP nearly disappeared under a planning-parameter-error scenario. Assuming R is not evidence that R exists.

Q3 · Blindness. [S/M] Without upstream wetland, the retention-present and retention-absent models have identical transitions for the specified unexposed conditions; the evaluated likelihood yields no Bayes-factor evidence there. **[E/M]** In development world d05, omitting R increased true U1 cost by **2.36**, while the omitting policy never occupied R-exposed states. In the explored development/stress worlds, there were **seven exact exposure traps across six worlds**. These are model results, not measurements from real catchments.

Q4 · Learning. [E/M] In d05 a planned probe produced retention exposure probabilities **0.77 / 0.95 / 0.99** at commitment years 1 / 2 / 3, but posterior movement probabilities **0 / 0 / 0.15**. **Exposure is not yet distinguishable evidence.** The exact feasible learner reached a maximum ratio to **fixed-probe oracle X*'s margin over the restricted fixed-policy menu B_extended **of** 0.107 (10.7%) **in the primary variant and** 0.179 (17.9%)* *in sensitivity; its declared screen was not met. Most apparent learner gain came from* state-contingent commitment, *not posterior updates. The oracle X* is constrained to the specified probe; it is the **comparison menu** that is called restricted.

Appendix A · Flood case: response-and-coupling matrix

These rows are **not** a single confirmatory five-arm comparison. The evidence labels must stay attached to each observation.

Response	Outcome / price if assumptions differ	Effect on what can be observed	Status / limitation
Keep: omit R	In d05 the true U1 cost increased by 2.36 when R mattered	Can retain zero exposure	**[E/M]** selected world
Change objective: COMP (F1-eco, omit R)	Better wetland outcomes; conventional U1 price	**[IP]** Wetland investment could expose R as a by-product; this coupling was **not tested as a learning intervention**	**[C/M]** package comparison; **[IP]** learning effect
Revise model: REP (assume R, U1)	Lower mean U1 than COMP under confirmatory conditions; less reliable advantage with parameter error	**[IP]** May reduce incentive to learn R for this decision; not tested as a complete feedback loop	**[C/M]** package result; **[E/M]** parameter error; **[IP]** learning effect
Probe / observe	Probe cost, delays and weak posterior gains in examined worlds	Exposes mechanism before evidence arrives	**[E/M]** selected worlds; **alternative sensor untested**
Hedge: MIX(R,q)	Equal-weighted mean U1 **6.465**–**6.638**, below pure F0 plans **6.924** (assume R) and **6.967** (omit R); wetland deficit between pure plans	Learning effect not established	**[E/M]** heuristic, *world means only*, not optimized belief-state planning

Comparison caution. MIX was evaluated as an exploratory per-step mixed-model heuristic, not as a confirmatory arm or optimal adaptive belief-state policy. The published T5 table compares MIX with **pure F0 commitments**, not every possible policy. Its wetland-deficit outcomes lay between those F0 commitments. Thus REP's confirmatory superiority to COMP on mean U1 never established REP as the best U1 policy in the study.

Coupling [IP]. Values influence wetland investment → investment determines whether R can operate in observable transitions → evidence may update beliefs → updated beliefs may alter later policies. **WP1 did not confirm this entire loop as one intervention.**

Provisional audit conclusion. The example warrants comparing multiple objectives, explicitly testing exposure and **time to informative evidence**, and benchmarking a feasible learner against simpler state-dependent and hedged rules. It does **not** select a universally best flood policy.

Status key. [C] confirmatory (missed precision) · [E] exploratory / development-world comparison · [S] exact property under the specified model · [M] model-derived · [IP] interpretation, not directly tested.

Appendix B · Hypothetical transfer: lending selection

Entirely fictional teaching example. No data, lender, statistical effect or intervention outcome is claimed here. It illustrates the structure of selective feedback; it is **not** a recommendation to experiment on applicants without safeguards.

Case card and decision relevance

An organization uses a model to approve or reject loan applications. It records repayment outcomes only for loans it actually issues. It is deciding whether to retain its threshold, revise its underwriting criteria, improve its measurement, or use a safeguarded learning strategy.

Gate: If rejected applicants' unobserved potential repayment outcomes would change the preferred approval policy, the missing counterfactual is decision-relevant. If every plausible model supports the same action, learning may have limited value—but **Q3 is still mandatory**.

The four probes

- **Q1 — Objective:** Does the organization assess only default risk and immediate revenue, or also fairness, access, long-term wellbeing and costs borne by applicants? Legitimate criteria and legal constraints must be specified, not guessed.
- **Q2 — Model:** Might risk predictions be inaccurate for applicants seldom approved? Could economic conditions, customer selection or unmeasured factors change the conclusion?
- **Q3 — Observability:** For rejected applicants, the counterfactual outcome *"repayment if this lender had approved"* is not directly observed. Approved-only outcomes can reinforce confidence in the selection rule. Outside credit records might inform some questions but do not automatically identify this loan-specific counterfactual.
- **Q4 — Learning:** Consider safer lawful options such as consent-based data collection, external validation, natural experiments or tightly governed trials. Ask when competing models would predict **different, observable outcomes**, who bears credit risk and whether a simpler conditional decision rule suffices. Random approvals are **not** presumed ethical or appropriate.

Response-and-coupling sketch

Response	Potential benefit	Missing evidence / risk
Keep current threshold	Operational continuity	Selective labels may persist
Revise objective	Makes excluded outcomes visible in deliberation	Could change who is approved and whose outcomes are observed
Revise model	May improve ranking with current evidence	Does not by itself reveal rejected applicants' counterfactual repayments
Observe / evaluate safely	Could improve decision-relevant evidence	Privacy, fairness, financial and consent/oversight duties
Robust / reversible hedge	May limit exposure to uncertain predictions	Could reduce the value of further learning, at a cost

Workshop prompt: Which people and outcomes are selected into the training and evaluation evidence? Which assumptions could the organization check without changing applicant treatment? What external oversight is required before any intervention? Who participates in defining an acceptable outcome? Which analogy with A2 fails? (One key difference is that financial outcomes involve human agency, law and potential harm; wetlands in A2 are synthetic state variables.)

v0.2.1 change record and evaluation plan

Revisions from v0.2: quick/full facilitation modes, concrete Q3 selective-evidence example and silent Q3 reflection, risk-bearer and safeguards fields, combined policy packages, model sensitivity example, decision-reframe disposition, and a separate after-action review. F0-C/MIX and oracle numerical statements are unchanged from v0.2.

Evidence of usefulness still needed: Pilot this frozen v0.2.1 against the team's ordinary decision process; record what consequential issues were *newly* surfaced, missed issues, burdens, time taken, what to cut, and decisions changed. A pleasant workshop is not proof of better decisions.

Optional companion · After-action blind-spot review

Use after the policy has operated long enough for meaningful observations. This is a one-page learning record, not evidence that the audit improved outcomes. **Do not infer unobserved counterfactual results from observed selected cases.**

Review field	Record
Decision, package, date adopted, reviewer, review date	_____
Expected outcomes / measurable predictions at decision time	_____
Actual observed outcomes; evidence status [O/M/H/U]	_____
Who was affected, who was consulted, unintended burdens or harms	_____
What did our policy select **into** the observed data?	_____
What groups, states or outcomes did it still prevent us from observing?	_____
Which assumptions were challenged; what remains unidentifiable?	_____
Did evidence arrive early enough to alter action? If not, why?	_____

Did the response make the decision more robust, or only make uncertainty harder to see? _____

What did we learn from outcomes, versus infer from a model or analogy? _____

Which uncertainty is still decision-relevant? Would resolving it change the action? _____

Safest feasible next step: ☐ Keep ☐ Adjust objective ☐ Revise model ☐ Gather evidence ☐ Hedge/combined ☐ Reframe ☐ Stop learning

Action, responsible person, safeguard, trigger and next review: _____

Disagreement / missing affected perspective to seek before next decision: _____

After-action restraint: If no credible counterfactual comparison exists, record the uncertainty. A policy may appear successful precisely because it suppresses observations that would challenge it. The review can identify that risk, **not prove the suppressed outcomes**.